

EDUCATION

The Chinese University of Hong Kong (CUHK)	Hong Kong SAR
<i>Doctor of Philosophy , Geophysics (Seismology)</i>	2026 – 2030
University of Tehran, Institute of Geophysics	Tehran, Iran
<i>Master of Science, Geophysics (Exploration Seismology)</i>	2025 – 2026
• Withdrew to accept Ph.D. fellowship	
Kharazmi University, Faculty of Earth Sciences	Tehran, Iran
<i>Bachelor of Science, Geology</i>	2021 – 2025
• Chairperson, Geological Society of Kharazmi University (June 2024 – June 2025)	
• Vice President of Media, Geological Society of Kharazmi University (November 2023 – June 2024)	
University of Lille, International Academy Lille Hdf	Lille, France
<i>Water Resource Management Summer School (5 ECTS)</i>	Summer 2024
• Presented a numerical groundwater modeling study on water abstraction impacts in the Weyib Watershed, Ethiopia.	

TEACHING EXPERIENCE

Iranian Young Scholars Club (IrYSC)	Tehran, Iran
<i>Earth Science Olympiad Teacher</i>	April 2021 – August 2026
Kharazmi University	Tehran, Iran
<i>Programming Teaching Assistant</i>	April 2025 – June 2025
• Programming Course, under supervision of Dr. Morteza Asemani	
Kharazmi University	Tehran, Iran
<i>Geostatistics Teaching Assistant</i>	February 2025 – June 2025
• Geostatistics Course, under supervision of Dr. Ehsan Pegah	
Young Scholars Club	Tehran, Iran
<i>Instructor</i>	July 2024
• Taught "Applied Data Science in Geology" for the Iran National IESO ¹ 2024 Team	
Kharazmi University	Tehran, Iran
<i>Instructor</i>	April 2024 – June 2024
• Organized and instructed the <u>Advanced Python and Machine Learning Workshop</u>	
NODET²	Tehran, Iran
<i>Teacher</i>	May 2021 – September 2021
• Taught ITEO ³ Marketing at Allameh Helli (Branches 5 & 8) and Farzanegan High Schools	

RESEARCH EXPERIENCE

FEZtool	Tehran, Iran
<i>Co-Founder & Research and Development Specialist</i>	August 2023 – Present
• Led Earth science research using advanced algorithms and geospatial analysis.	
International Astronomical Search Collaboration, NASA Partner	Remote
<i>Asteroid Search Team Member</i>	July 2025 – September 2025
• Analyzed Pan-STARRS data, contributing to 3 preliminary asteroid discoveries.	
Kharazmi University	Tehran, Iran
<i>Research Assistant</i>	October 2024 – January 2025
• Performed MASW ⁴ to extract dispersion images under supervision of Dr. Mehdi Talkhablou.	

COMMUNITY SERVICE

Zamin Pouya Scientific Journal Vol. 31	Tehran, Iran
<i>Associate Editor & Reviewer</i>	August 2025 – June 2026
The 4th National Festival of Young Iranian Geologists	Tehran, Iran
<i>Education Committee Member</i>	April 2025 – May 2025

¹International Earth Science Olympiad

²National Organization for Development of Exceptional Talents

³Iranian Thinking & Entrepreneurship Olympiad

⁴Multichannel Analysis of Surface Waves

SELECTED PUBLICATIONS

Journal Articles

1. Talkhablou, M., **Farmahini Farahani, M.**, & Mansouri, S. S. (2025). A Python-Based Framework for Land Cover Classification in Engineering Geology: A Comparative Assessment of SVM, K-Means, and Spectral Indices. *Kharazmi Journal of Earth Sciences*.
2. **Farmahini Farahani, M.** (2024). Designing the green belt in the southwest of Zahedan. *Ovan Scientific Journal*
3. Saravani, M. J., Kashef, S., **Farmahini Farahani, M.**, Kashef, M., & Zohreh, M. (2023). Investigating the accuracy of hybrid models with wavelet transform in the forecast of watershed runoff. *Journal of Water Management Modeling (JWMM)*

Conference Papers

1. **Farmahini Farahani, M.** (2025, August). Machine Learning-Enhanced Seismic Event Detection and Noise Reduction for Planetary Seismology: Insights from Mars and Lunar Data. In *Seismology Student Workshop (SSW 2025)*, GatherTown (Virtual)
2. Rezaei, K., & **Farmahini Farahani, M.** (2024, March). Visualization of magnetometric data with Python (case study: Segregation sedimentary units of Torud Playa). In *The 1st Conference on Applied Geophysics in Mines*, Tehran, Iran
3. Rezaei, K., & **Farmahini Farahani, M.** (2024, February). Comparing the performance of the k-means clustering method and spectral indices in characterizing land use and land cover using remote sensing with Python (case study: Landsat 8 multi-spectral images). In *2nd International Conference for Iranian Geography and Earth Sciences Students*, Tehran, Iran
4. **Farmahini Farahani, M.**, Rezaei, K., Karamollahi, S., & Moghadam, F. J. (2024, January). Application of PCA algorithm in remote sensing using Python programming language (case study: Landsat 9 multispectral images). In *8th Symposium of Sedimentological Society of Iran*, Hormozgan, Iran
5. Rezaei, K., & **Farmahini Farahani, M.** (2023, September). The importance of using Python programming language in Earth science. In *26th Conference of Geological Society of Iran*, Urmia, Iran

Books

1. **Farmahini Farahani, M.**, Najibifar, Y., & Akbarzadeh Afshari, A. (2025). *Earthquake Prediction Using Geophysical Data Analysis and Artificial Intelligence*. **Heritage Branch, Library and Archives Canada**
2. Rezaei, K., **Farmahini Farahani, M.**, Karimzadehasl, E. (Trans.). (2024). *Introduction to Python in Earth science data analysis: From descriptive statistics to machine learning*. **Tasir Publication**. (Original work published by Maurizio Petrelli)

PROJECTS

PlanetaryGeophysics | github.com/aradfarahani/PlanetaryGeophysics 2025 – Present

- Built an open-source Python platform for analyzing and visualizing planetary geophysical data.

FEZrs (Open Source) | pypi.org/project/fezrs 2023 – Present

- Advanced Python library for remote sensing, developed by FEZtool with over 100,000 downloads

Seismicity Analysis in Iran (2010–2025) | iris.edu/hq/workshops/2025/06 2025

- performed Iran seismicity analysis for the *Seismology Skill-Building Workshop 2025* final project.

Goelectricspy (Open Source) | zenodo.org/records/10966815 2023

- Interactive 3D visualization tool for subsurface resistivity using geoelectrics survey data

INVITED TALKS

1. **Walrus: A Cross-Domain Foundation Model for Continuum Dynamics:** *Applied ML Lab, Center for Complex Networks, Department of Physics, Shahid Beheshti University*; 13 December 2025

2. **Applied Artificial Intelligence in Earth Science:** *Faculty of Geosciences, Shahrood University of Technology & Faculty of Earth Sciences, Kharazmi University*; 11 November 2025
3. **Applications of Artificial Intelligence in Education and Earth Science Analysis:** *Educational Technology Expert Group, Karaj and Fardis County Department of Education, District 2*; 30 December 2024

LICENSES & CERTIFICATIONS

Applied Geotechnical Seismology , Kharazmi University	2026
Applications of Non-Destructive Geophysical Methods in Engineering Geology Studies , Kharazmi University	
Seismology Skill Building Workshop 2025 , EarthScope Consortium	2025
Investigating practical issues in the exploration of Sulphide Deposits using the IP-RS method , National Iranian Geophysical Society	
Remote Pilot Certification (Multirotor) , Civil Aviation Authority of Islamic Republic of Iran	2023
Unmanned Aerial Vehicle Surveying , Maan Academy	
GIS, Mapping, and Spatial Analysis Specialization , University of Toronto (Offered by Coursera)	
CS50 Certificate , CS50 (Harvard University)	2021

HONORS & AWARDS

CUHK's International Student PhD Fellowship - <i>Awardee</i> , CUHK	2026
9th National Young Scientists Festival - <i>Merit Award Winner</i> , <i>Geology Category</i>	2025
Hydro 2024 Research-Oriented Summer School - <i>Full Scholarship</i> , University of Lille	2024
Research Visit Grant (2500 DKK) - <i>Awardee</i> , University of Copenhagen	2022
DrCT 2021 Global Computational Competition - <i>Gold Medalist</i>	2021
1st national Thinking & Entrepreneurship Olympiad (ITEO) - <i>Silver Medalist</i>	2020
WMI 2015 World Mathematical Championships Selection - <i>Silver Medalist</i>	2015

SKILLS & INTERESTS

Programming Languages: Python (ObsPy, pyGIMLi, Astropy, OpenCV, TensorFlow, PyGMT, Xarray, Scikit-learn, kepler.gl, Leafmap, Folium, Rasterio, GeoPandas), Julia, Bash (Shell scripting), SQL (MySQL), GMT, GDAL; Familiar with R

Markup Languages: Markdown, L^AT_EX

Softwares: Astrometrica, ENVI, Seismic Analysis Code(SAC), SeisAn, QGIS, SeisImager, Blender, ilastik, Google Earth Pro, Refrapy, SAS.Planet, draw.io, Groundwater Vistas

Field Techniques: Drone Surveying, Geoelectrical Methods (Induced Polarization–Resistivity Sounding, IP-RS), Seismic Refraction Data Acquisition

Languages: Farsi (Persian): *Native*; English: *Fluent (C1)*; German: *Elementary (A2)*; Mandarin: *Beginner (Duolingo Score: 10)*

Research Interests: Signal/Image Processing, Geophysics (Seismology), Remote Sensing, Machine Learning, Open-Source Software Development

PROFESSIONAL MEMBERSHIPS

National Iranian Geophysical Society	Student member, 2025 – Present
Space Generation Advisory Council	Member, 2025 – Present
Google Maps Platform Innovators	Member, 2025 – Present
Geological Society of Iran	Student member, 2021 – 2022